AI-CAD House Builder: A Tool-Using Large Language Model Framework for Natural-Language Architectural CAD

Shyamsundar Kamalakannan[1], Dhasaradhan D[2], Prajeshwaran B[3], G. Revathy[4]
[4]Associate Professor,[1,2,3]UG Scholar
Department of Artificial Intelligence and Machine Learning, Vels Institute of Technology Pallavaram ,India
Email id: sundarshyam5780@gmail.com,prajeshpraju2006@gmail.com,
dhasaradhandevaranjith@gmail.com , revathy.se@vistas.ac.in

Abstract

Architectural computer-aided design (CAD) usually requires users to understand specialized commands, coordinates, dimensions, and the correct order of modeling steps. These requirements can make house design difficult for people without CAD experience. This paper introduces AI-CAD House Builder, a framework that allows users to create and modify residential CAD models by describing their needs in everyday language. It follows a planner, tool, validation, and geometry workflow, keeping language interpretation separate from the actual construction of shapes. A locally run Qwen2.5 7B language model, served using llama.cpp, interprets instructions and selects operations for adding rooms, walls, doors, and windows, as well as moving, resizing, or removing objects. Before an operation is carried out, a validation layer checks its parameters, referenced objects, and basic geometric constraints. A deterministic CAD engine then performs the approved operation. The application combines a web-based 2D design workspace with interactive 3D viewing, while a FastAPI backend maintains the model state. Prototype tests show that natural-language instructions can be converted into structured CAD actions and that users can make follow-up changes without rebuilding the entire design. The key idea is to use the language model for understanding and planning, while relying on deterministic software to validate commands and construct geometry. This approach offers a more controllable and privacy-conscious way to make architectural CAD easier to use.

Keywords— Large Language Models, Natural-Language CAD, Tool-Using LLM, Architectural Design, Qwen2.5, llama.cpp